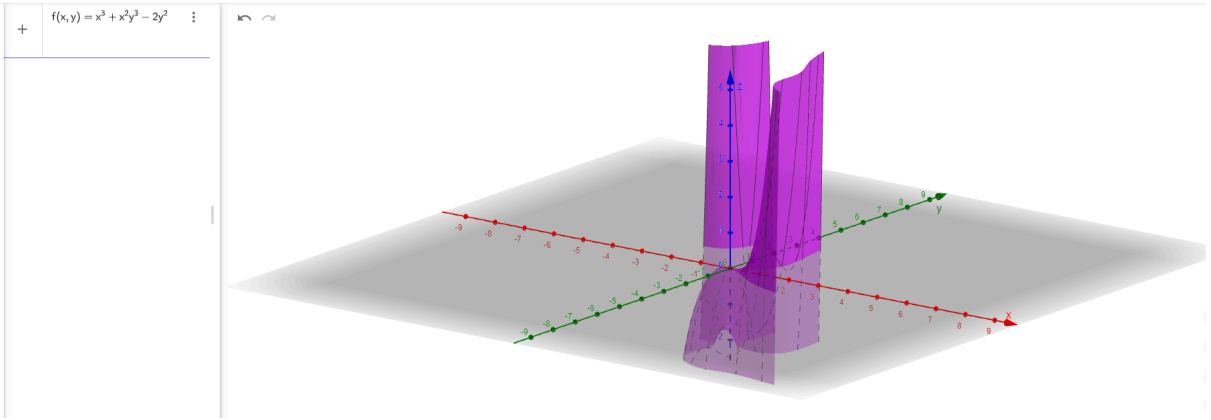
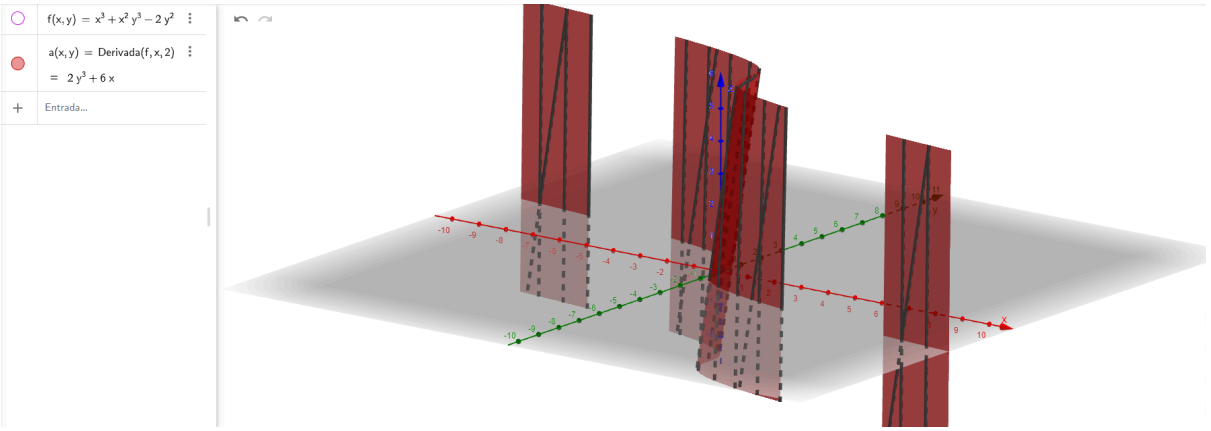
Medida de Eficiência - Cálculo II

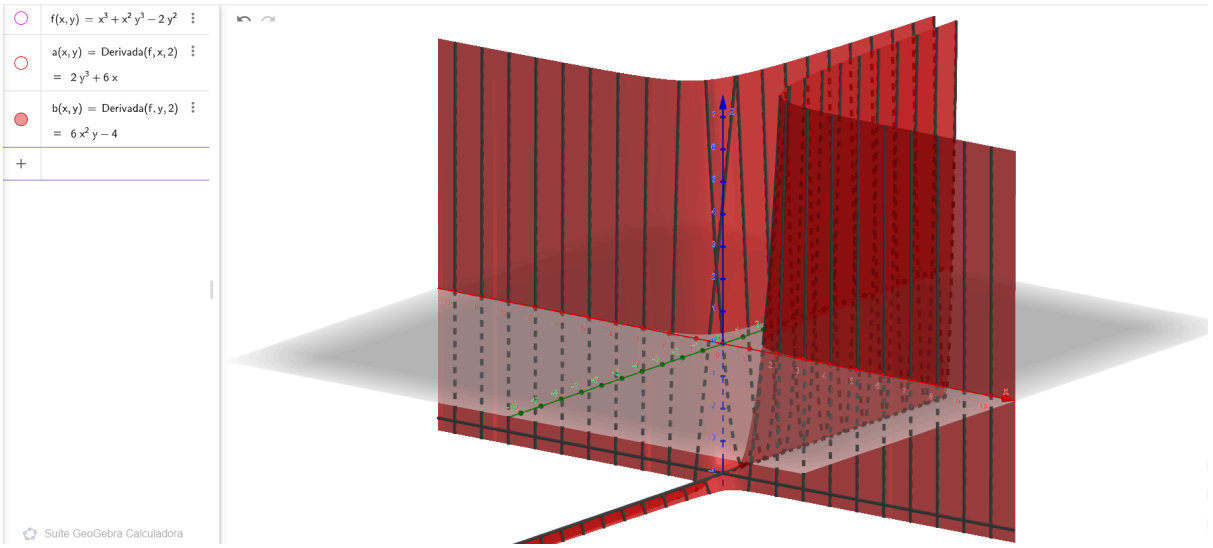
Aluno: Francisco Lavres Correia da Fonseca

Professor: Diego Melo Costa

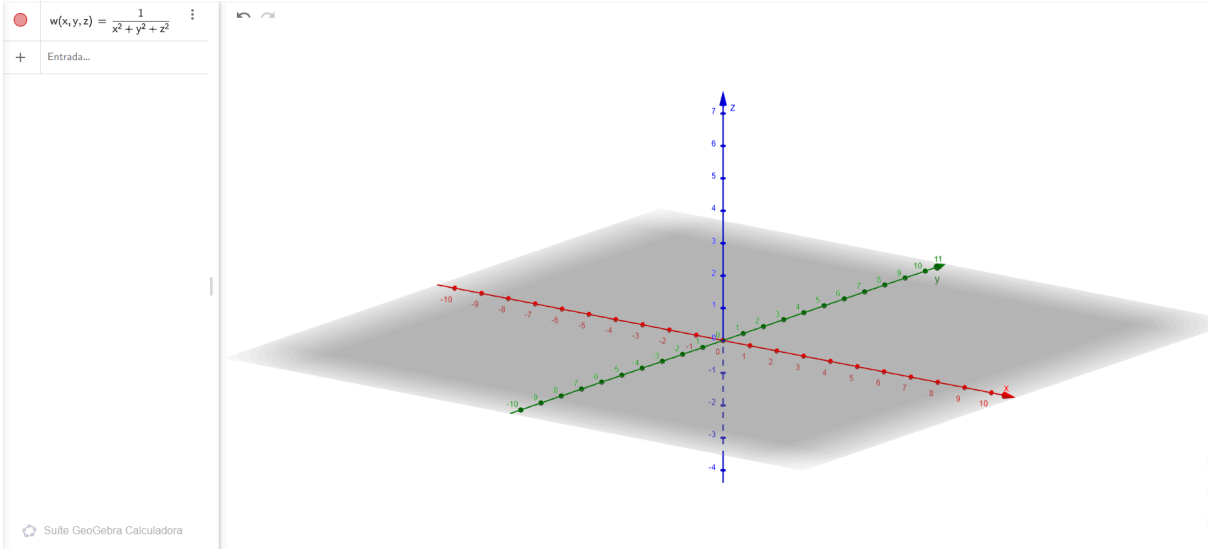
1ª) a)

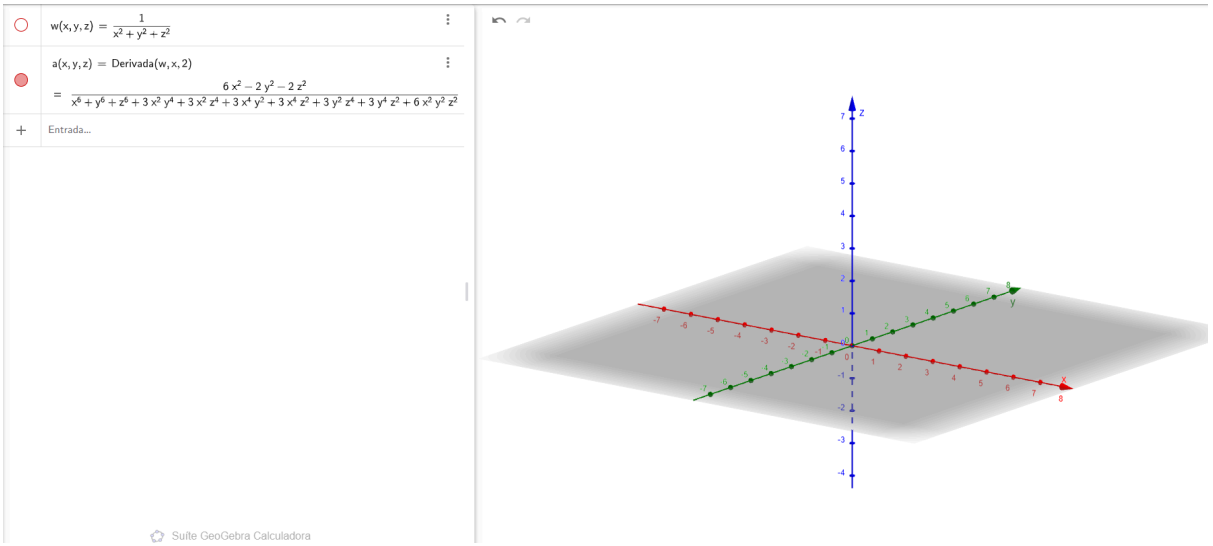


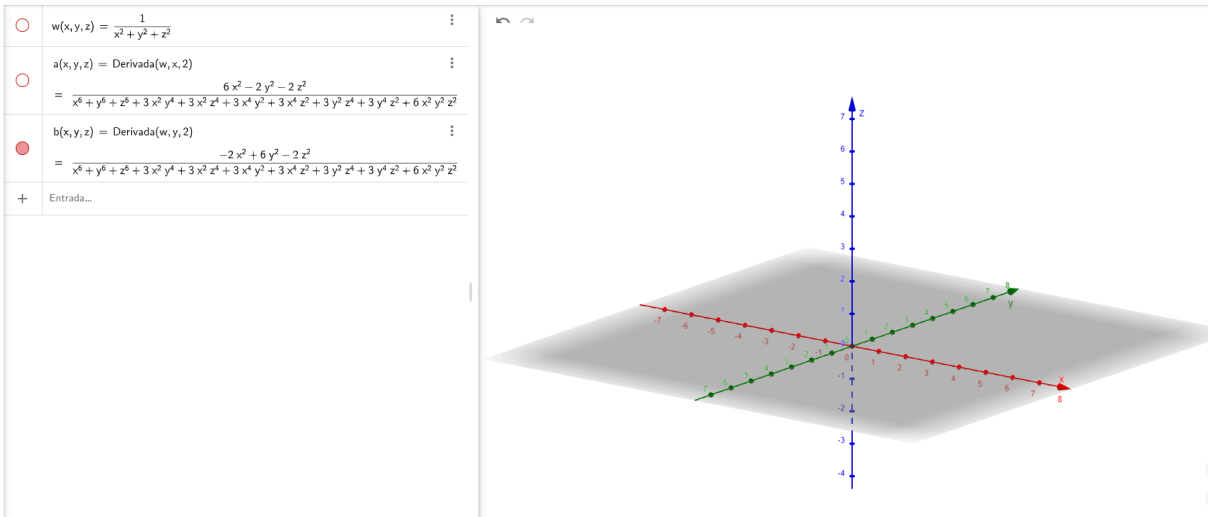


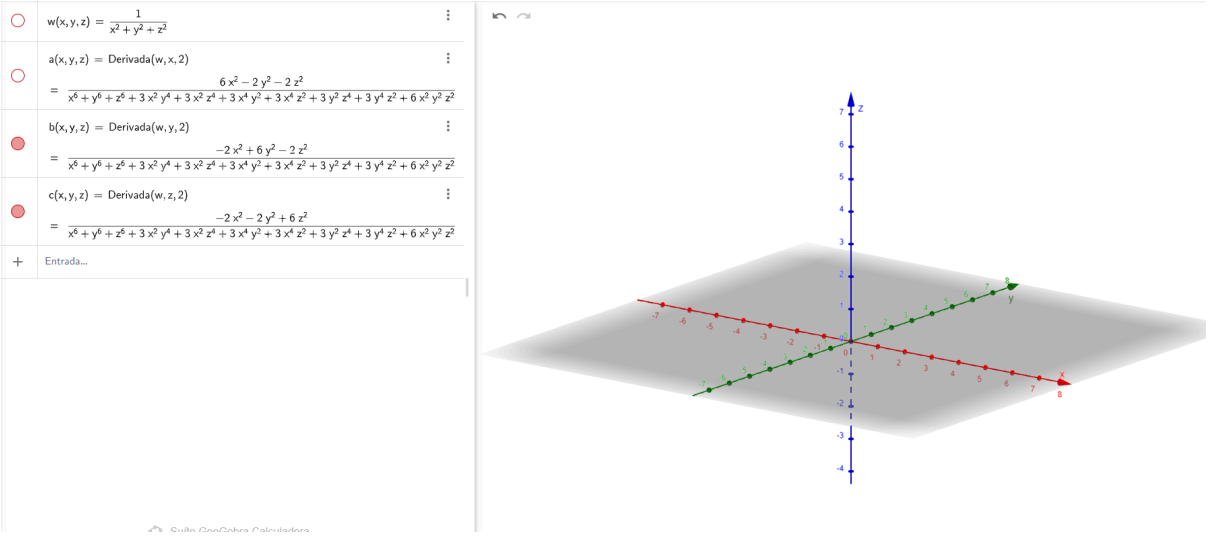


b)

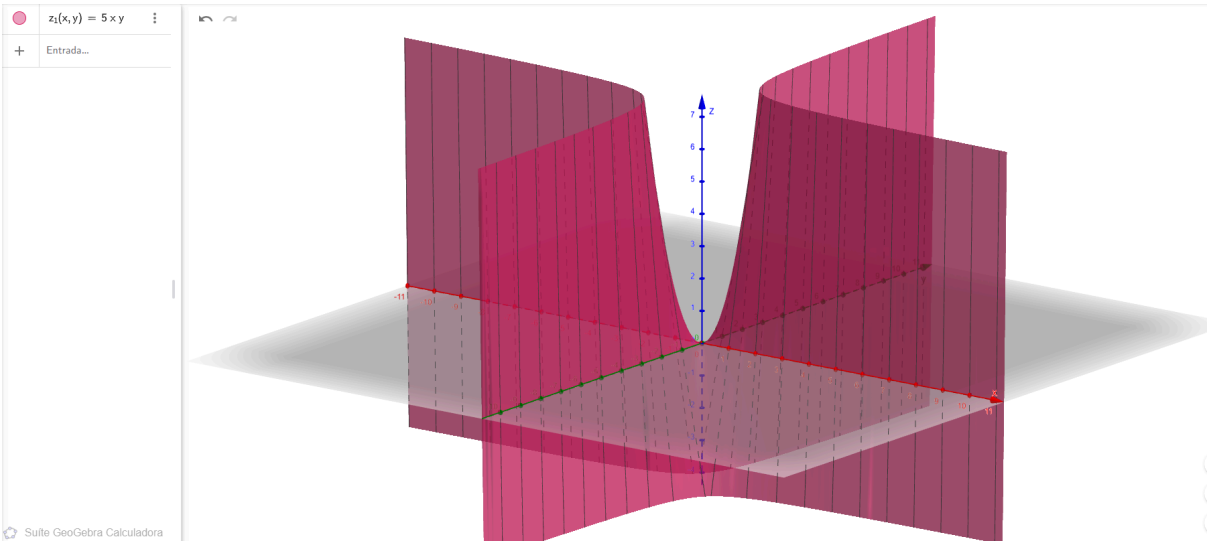

$w(x, y, z) = \frac{1}{x^2 + y^2 + z^2}$
Entrada...
Suite GeoGebra Calculadora

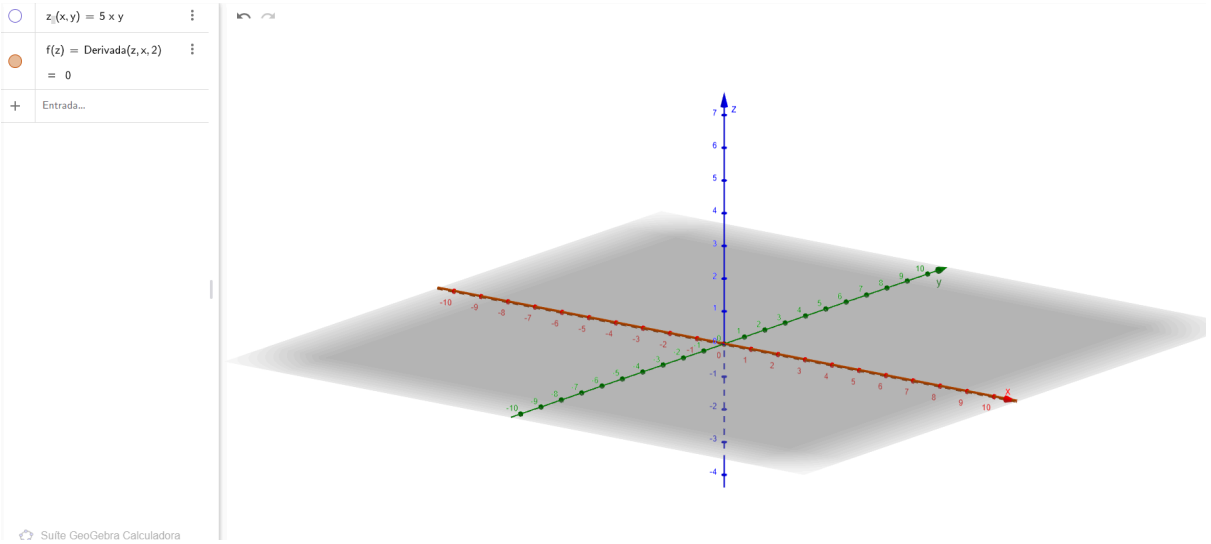

$w(x, y, z) = \frac{1}{x^2 + y^2 + z^2}$
$a(x, y, z) = \text{Derivada}(w, x, 2)$
$= \frac{6x^2 - 2y^2 - 2z^2}{x^6 + y^6 + z^6 + 3x^2y^4 + 3x^2z^4 + 3x^4y^2 + 3x^4z^2 + 3y^2z^4 + 3y^4z^2 + 6x^2y^2z^2}$
Entrada...
Suite GeoGebra Calculadora

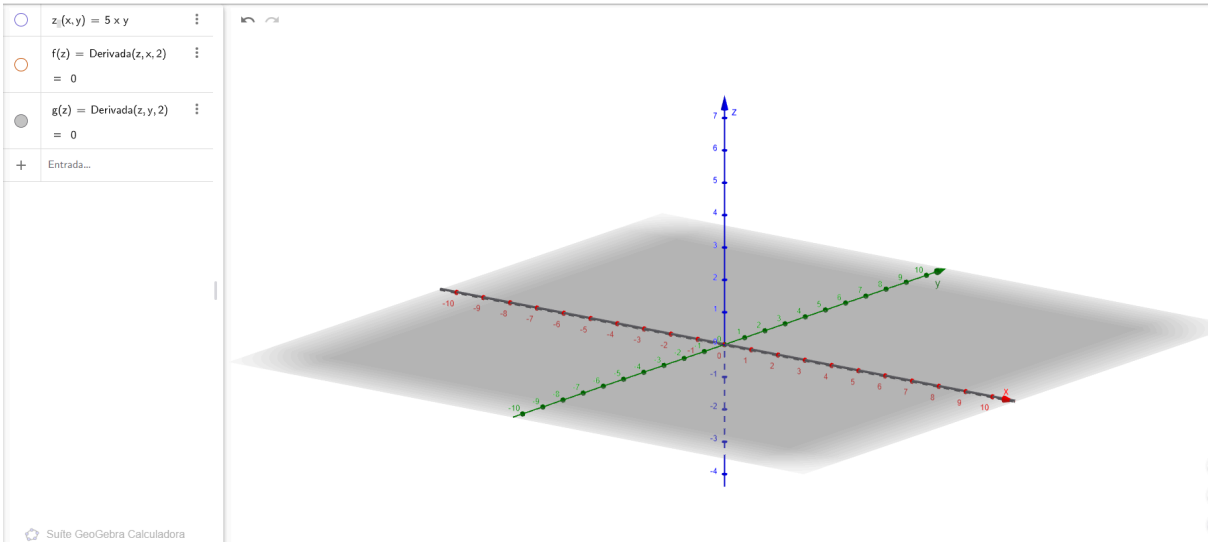

$w(x, y, z) = \frac{1}{x^2 + y^2 + z^2}$
$a(x, y, z) = \text{Derivada}(w, x, 2)$
$= \frac{6x^2 - 2y^2 - 2z^2}{x^6 + y^6 + z^6 + 3x^2y^4 + 3x^2z^4 + 3x^4y^2 + 3x^4z^2 + 3y^2z^4 + 3y^4z^2 + 6x^2y^2z^2}$
$b(x, y, z) = \text{Derivada}(w, y, 2)$
$= \frac{-2x^2 + 6y^2 - 2z^2}{x^6 + y^6 + z^6 + 3x^2y^4 + 3x^2z^4 + 3x^4y^2 + 3x^4z^2 + 3y^2z^4 + 3y^4z^2 + 6x^2y^2z^2}$
Entrada...

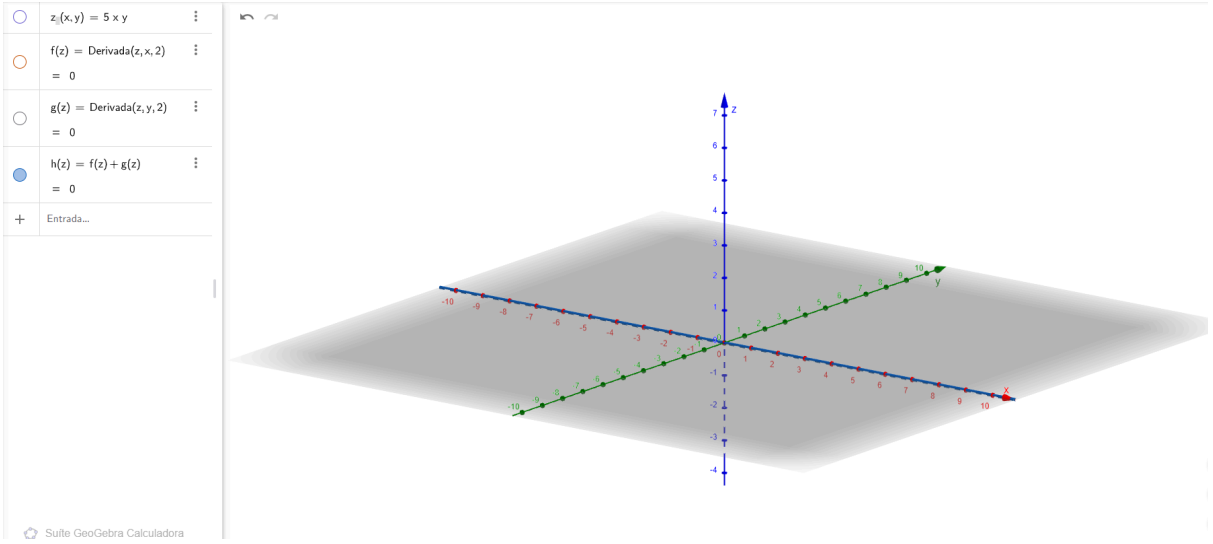

w(x, y, z) = 1 / (x² + y² + z²)
a(x, y, z) = Derivada(w, x, 2)
= (6 x² − 2 y² − 2 z²) / (x⁶ + y⁶ + z⁶ + 3 x² y⁴ + 3 x² z⁴ + 3 x⁴ y² + 3 x⁴ z² + 3 y² z⁴ + 3 y⁴ z² + 6 x² y² z²)
b(x, y, z) = Derivada(w, y, 2)
= (−2 x² + 6 y² − 2 z²) / (x⁶ + y⁶ + z⁶ + 3 x² y⁴ + 3 x² z⁴ + 3 x⁴ y² + 3 x⁴ z² + 3 y² z⁴ + 3 y⁴ z² + 6 x² y² z²)
c(x, y, z) = Derivada(w, z, 2)
= (−2 x² − 2 y² + 6 z²) / (x⁶ + y⁶ + z⁶ + 3 x² y⁴ + 3 x² z⁴ + 3 x⁴ y² + 3 x⁴ z² + 3 y² z⁴ + 3 y⁴ z² + 6 x² y² z²)
Entrada...

2ª)a)


z₁(x, y) = 5 x y
Entrada...
Suite GeoGebra Calculadora

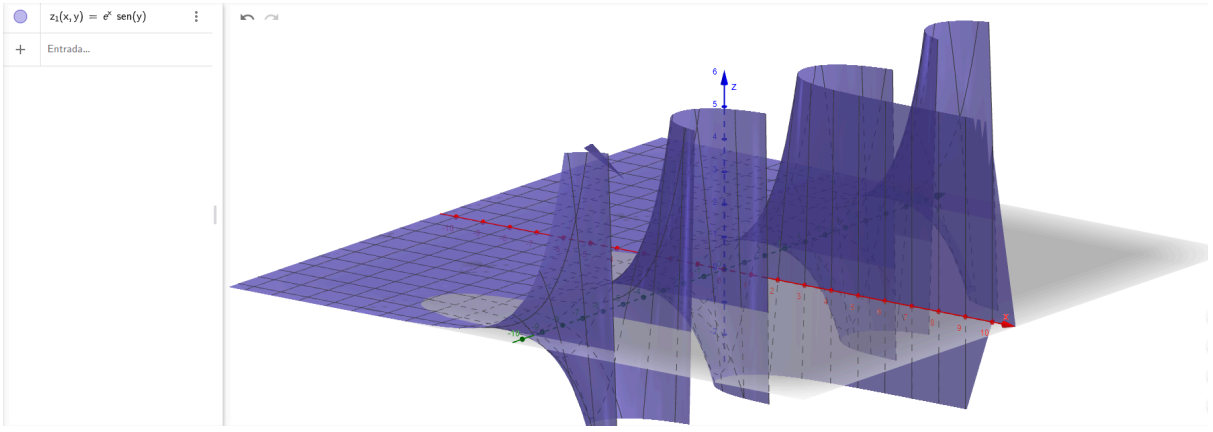

z₁(x, y) = 5 x y
f(z) = Derivada(z, x, 2)
= 0
Entrada...
Suite GeoGebra Calculadora

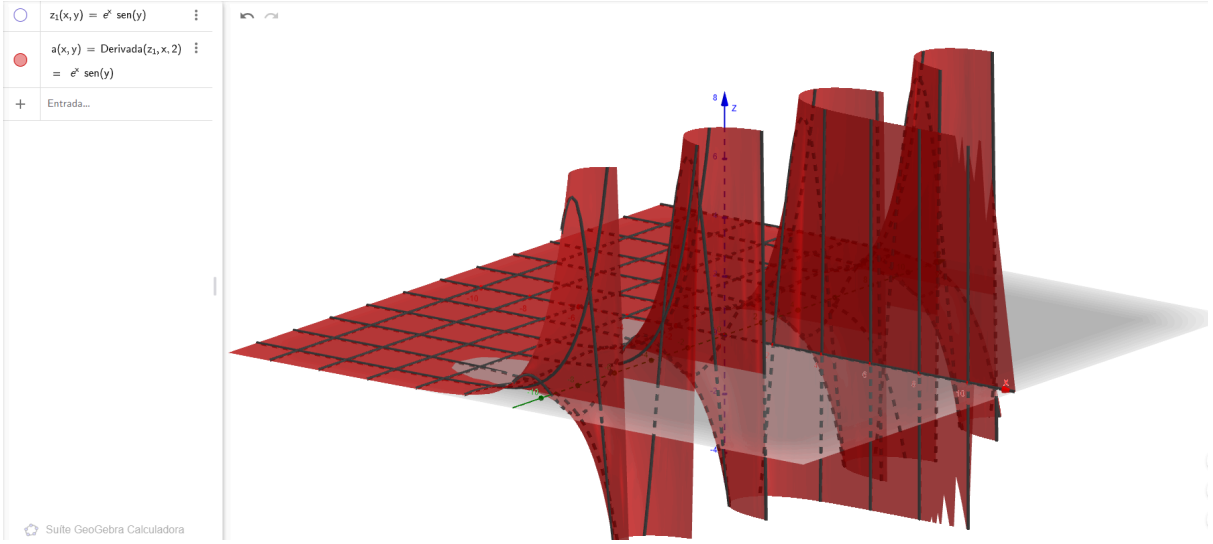


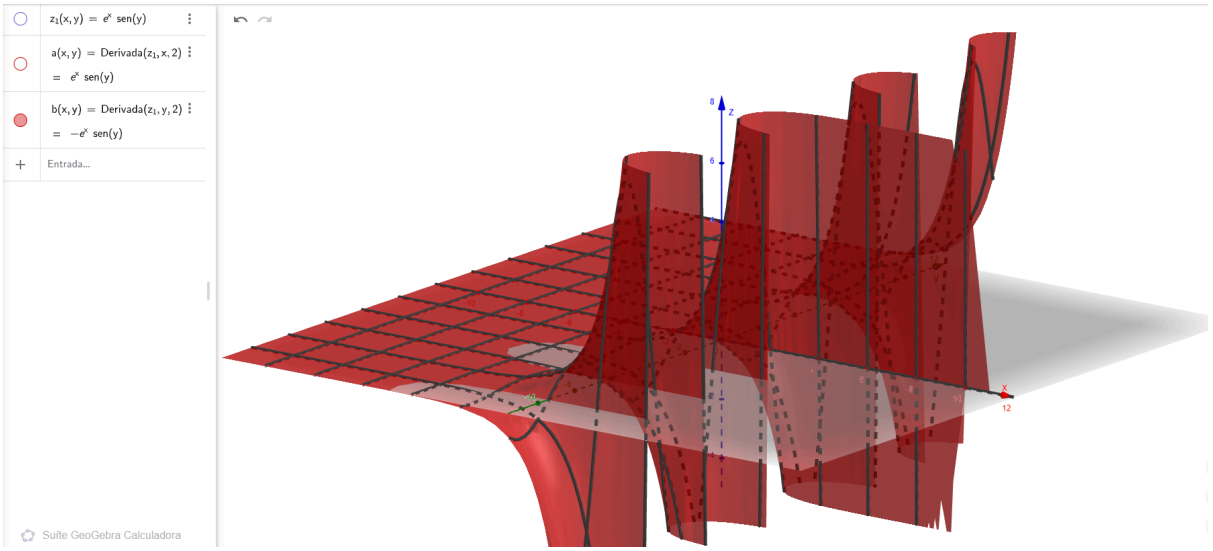


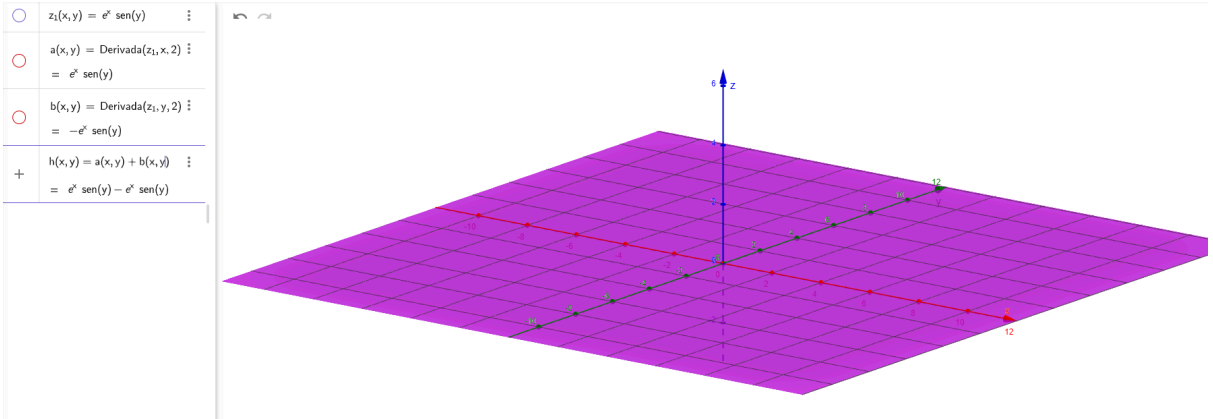
SATISFAZ A EQUAÇÃO

b)



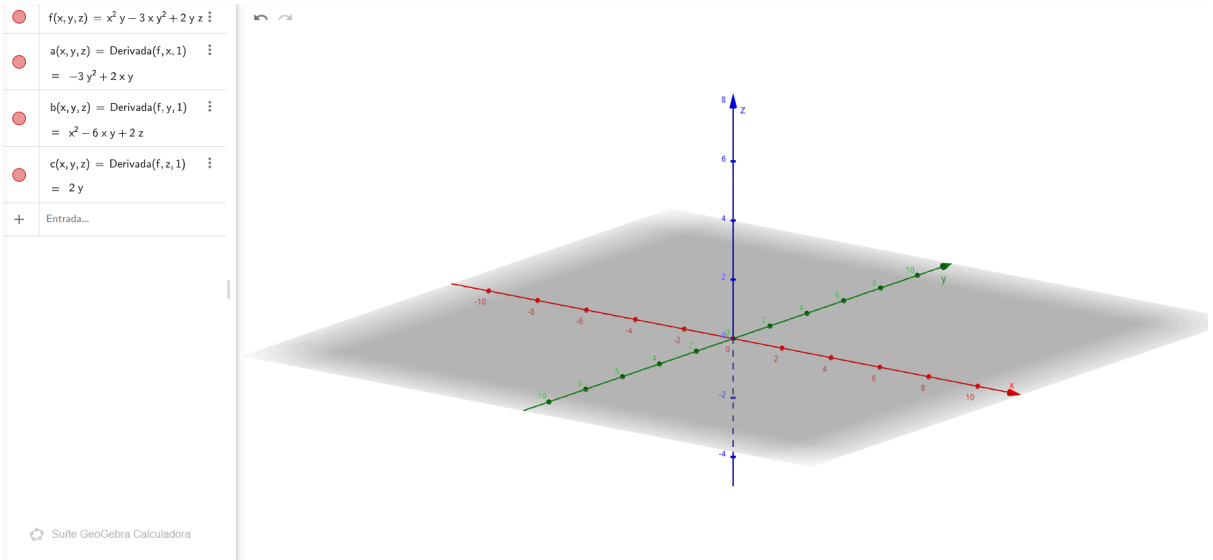




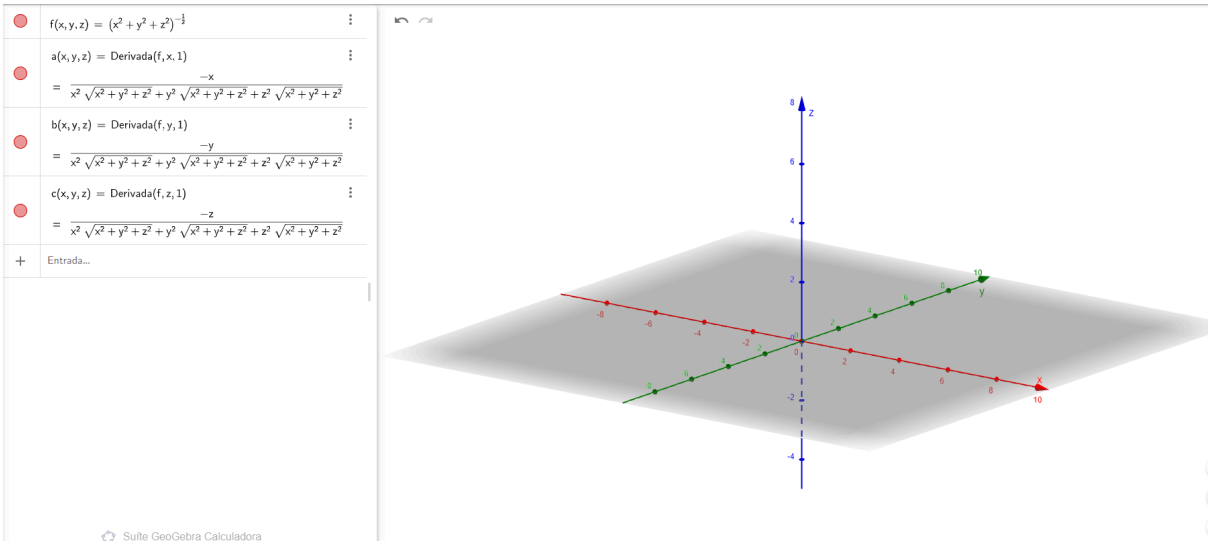


SATISFAZ A EQUAÇÃO

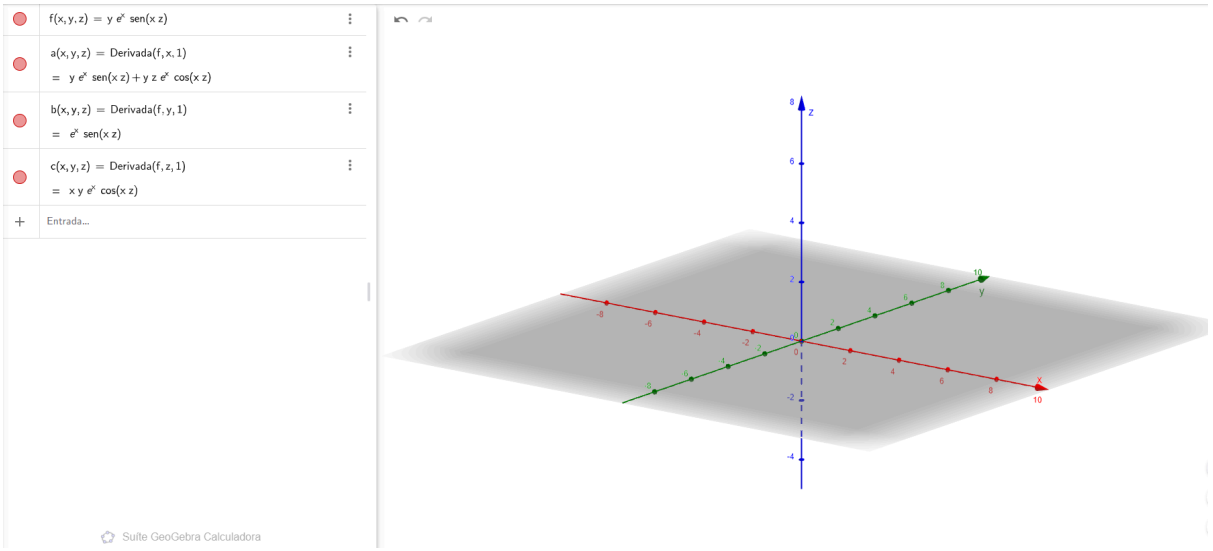
3ª) a)


f(x, y, z) = x² y − 3 x y² + 2 y z
a(x, y, z) = Derivada(f, x, 1)
= −3 y² + 2 x y
b(x, y, z) = Derivada(f, y, 1)
= x² − 6 x y + 2 z
c(x, y, z) = Derivada(f, z, 1)
= 2 y
Entrada...
Suite GeoGebra Calculadora

b)


f(x, y, z) = (x² + y² + z²)^(−1/2)
a(x, y, z) = Derivada(f, x, 1)
= −x / (x² √(x² + y² + z²) + y² √(x² + y² + z²) + z² √(x² + y² + z²))
b(x, y, z) = Derivada(f, y, 1)
= −y / (x² √(x² + y² + z²) + y² √(x² + y² + z²) + z² √(x² + y² + z²))
c(x, y, z) = Derivada(f, z, 1)
= −z / (x² √(x² + y² + z²) + y² √(x² + y² + z²) + z² √(x² + y² + z²))
Entrada...
Suite GeoGebra Calculadora

c)


f(x, y, z) = y e^x sen(x z)
a(x, y, z) = Derivada(f, x, 1)
= y e^x sen(x z) + y z e^x cos(x z)
b(x, y, z) = Derivada(f, y, 1)
= e^x sen(x z)
c(x, y, z) = Derivada(f, z, 1)
= x y e^x cos(x z)
Entrada...
Suite GeoGebra Calculadora
z
y
x